

Average Quantities

$$\underbrace{\bar{h} = \frac{\bar{q''}}{\Delta T}}_{T_0 = \text{constant}} \quad \text{or} \quad \underbrace{\bar{h} = \frac{q''}{\Delta T}}_{q''|_{y=0} = \text{constant}}$$

For $T_0 = \text{constant}$:

$$\bar{h} = \frac{1}{\Delta T} \left[\frac{1}{L} \int_0^L q'' dx \right] = \frac{1}{L} \int_0^L h dx$$

For $q''|_{y=0} = \text{constant}$:

$$\bar{h} = \frac{q''|_{y=0}}{\frac{1}{L} \int_0^L \Delta T dx}$$

$$\text{The average } Nu_L = \overline{Nu}_L = \frac{\bar{h}L}{k} \neq \frac{1}{L} \int_0^L Nu_x dx$$

For a flat plate:

$$\bar{h} = \frac{1}{L} \int_0^L h dx \Rightarrow h = \frac{k}{x} Nu_x$$

$$\bar{h} = \frac{0.332 k Pr^{1/3}}{L} \sqrt{\frac{U_\infty}{\nu}} \int_0^L \frac{x^{1/2}}{x} dx \Rightarrow \boxed{\bar{h} = 0.664 Re_L^{1/2} Pr^{1/3} \cdot \frac{k}{L}}$$

$\hookrightarrow T_0 = \text{constant}, Pr > 0.5$

$$\boxed{\overline{Nu}_L = \frac{\bar{h}L}{k} = 0.664 Re_L^{1/2} Pr^{1/3}} \quad Pr > 0.5$$

$$\boxed{Nu_L = 1.13 Re_L^{1/2} Pr^{1/2}} \quad Pr < 0.5$$

Some Observations and Notes

So far, our results are valid for the following conditions

- 1) $Re_x = \frac{\rho U_\infty x}{\mu}$ or $Re_L = \frac{\rho U_\infty L}{\mu} < 5.0 \times 10^5$ (Laminar)
- 2) $Ma = \frac{U_\infty}{\text{sound speed}} < 0.3$ (Incompressible)
- 3) $Ec \equiv \text{Eckert Number} = \frac{U_\infty^2}{c_p(T_0 - T_\infty)} \ll 1 \Rightarrow$ (Viscous dissipation heating is negligible)
- 4) Evaluate fluid properties at the b.l. film temp.:

$$T_f = \frac{T_0 + T_\infty}{2}$$

*5) $h_x \sim \frac{1}{x^{1/2}} \Rightarrow \text{As } x \rightarrow 0, h_x \rightarrow \infty$

The boundary layer model breaks down in the region of $x=0$.

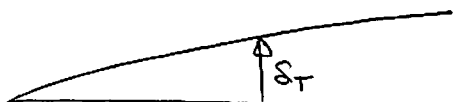
6) So far, we've only dealt with $T_0 = \text{constant}$. We will solve for the $q_{y=0}'' = \text{constant}$ case later, with the integral technique.

* Note although the b.l. solution diverges at $x \rightarrow 0$, in real life, h_x is actually higher in this region. Hence it's more beneficial to re-start your b.l. as often as possible to minimize δ_T and maximize entrance effects. We will discuss these later in the class.

i.e.



$\Rightarrow$ Good, $h \uparrow$, however $\tau \uparrow$ as well



$\Rightarrow$ Bad, $h \downarrow$, however $\tau \downarrow$ as well

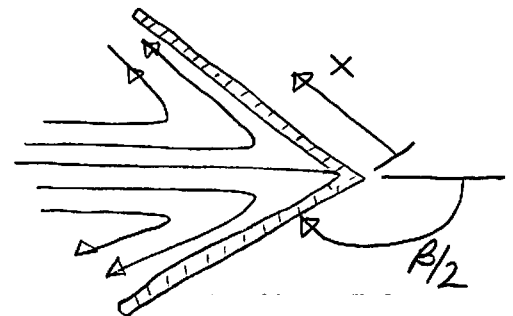
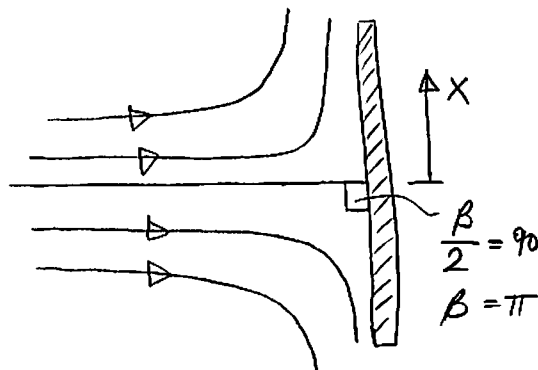
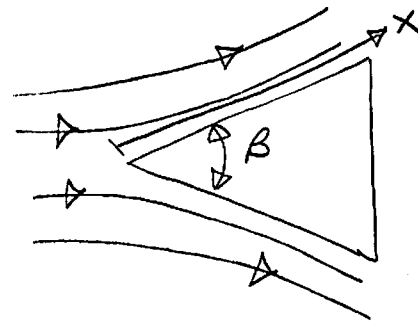
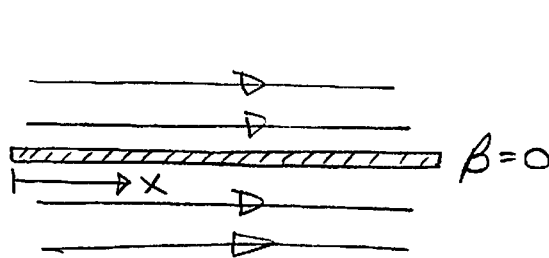
Similarity Solutions for Flow with Longitudinal Pressure Grad.
 So how do we deal with flows which have longitudinal pressure gradients?

It turns out there is a class of similarity solutions that work for potential flow problems, in 2D.

For 2D potential flows:

$$U_{\infty}(x) = Cx^m \Rightarrow \text{For a derivation of this, visit a potential flow fluids textbook.}$$

$$m = \frac{\beta}{2\pi - \beta} = \frac{x}{U_{\infty}} \cdot \frac{dU_{\infty}}{dx}$$



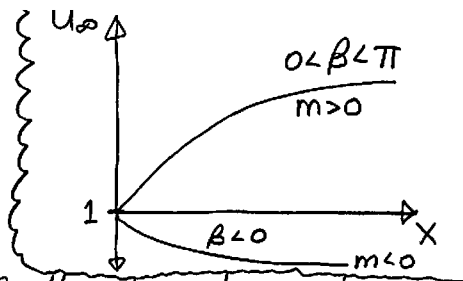
Note for these cases, we cannot assume $\frac{\partial P}{\partial x} = 0$, except for $\beta = 0$. The varying cross section in each flow induces a pressure change in the longitudinal direction.

Also note, if $\beta < 0$, this means:

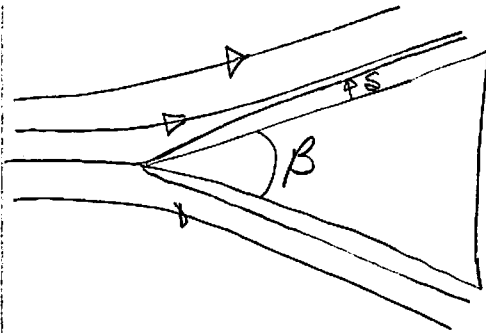


Our b.l. equation (momentum) becomes:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \frac{\partial^2 u}{\partial y^2} \quad (1)$$



Taking a streamline at the edge of the boundary layer, or within a b.l. reveals: (note $\partial p / \partial y \ll \partial p / \partial x$ still)



$$P_\infty + \frac{1}{2} \rho U_\infty^2 = \text{constant}$$

$$\frac{\partial P_\infty}{\partial x} + \rho U_\infty \frac{\partial U_\infty}{\partial x} = 0$$

$$-\frac{\partial P}{\partial x} = \rho U_\infty \frac{\partial U_\infty}{\partial x} \quad (2)$$

Back substituting (2) into (1), we obtain:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = U_\infty \frac{\partial U_\infty}{\partial x} + \nu \frac{\partial^2 u}{\partial y^2} \quad (3)$$

We know $U_\infty = Cx^m \Rightarrow U_\infty \frac{\partial U_\infty}{\partial x} = Cx^m m Cx^{m-1}$

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \frac{U_\infty^2 m}{x} + \nu \frac{\partial^2 u}{\partial y^2} \quad (4)$$

To solve equation (4) we can actually use the identical similarity variable we defined earlier.

$$\eta = y \sqrt{\frac{U_\infty}{\nu x}}$$

Instead of using the same procedure as before (i.e. defining $f' = u/U_\infty$, & solving for $u, v = f(f', \eta)$). We can define a streamfunction $\psi(x, y)$:

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

To check if this works, we can check continuity

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\frac{\partial}{\partial x} \frac{\partial \psi}{\partial y} - \frac{\partial}{\partial y} \frac{\partial \psi}{\partial x} = 0 \Rightarrow \text{Satisfies continuity (Note, only valid for analytic functions)}$$

Now we can do one more thing;

$$\frac{\partial^2 g}{\partial x \partial y} - \frac{\partial^2 g}{\partial y \partial x} = 0$$

$$f' = \frac{u}{u_\infty} \Rightarrow u = \frac{\partial \psi}{\partial y}$$

$$f' = \frac{1}{u_\infty} \cdot \frac{\partial \psi}{\partial y}$$

$$u_\infty f' = \frac{\partial \psi}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \Rightarrow \eta = y \sqrt{\frac{u_\infty}{\nu x}} \Rightarrow \frac{\partial \eta}{\partial y} = \sqrt{\frac{u_\infty}{\nu x}}$$

$$u_\infty^{1/2} f' \frac{\sqrt{\nu x}}{\sqrt{u_\infty}} = \frac{\partial \psi}{\partial \eta}$$

$$\psi = \sqrt{u_\infty \nu x} \int f' d\eta \Rightarrow \boxed{\psi = (u_\infty \nu x)^{1/2} f} \quad (5)$$

Now we can transform our b.l. equation

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \frac{u_\infty^2 m}{x} + \nu \frac{\partial^2 u}{\partial y^2} \quad (6)$$

Back substitute (5) and $u = \frac{\partial \psi}{\partial y}$, $v = -\int \frac{\partial u}{\partial x} dy$ into (6) Continuity $u_\infty = f(x)$

Also, you will need:

$$\frac{\partial \psi}{\partial x} = \frac{\partial \psi}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} \Rightarrow \frac{\partial \eta}{\partial x} = \frac{\partial}{\partial x} \left(y \sqrt{\frac{u_\infty}{\nu x}} \right)$$

$$\frac{\partial \psi}{\partial y} = \frac{\partial \psi}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \Rightarrow \frac{\partial \eta}{\partial y} = \sqrt{\frac{u_\infty}{\nu x}}$$

$$\frac{\partial \psi}{\partial \eta} = (u_\infty \nu x)^{1/2} f'; \text{ remember } u_\infty = Cx^m$$

Simplifying, we obtain: $\boxed{f''' + \frac{1}{2}(m+1)ff'' + m(1-f')^2 = 0}$

↳ For full derivation, see pg. 64A-64D

Extra Derivation: Falkner-Skan Momentum Equation

We want to go from PDE to ODE:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \frac{U_\infty^2 m}{x} + \nu \frac{\partial^2 u}{\partial y^2} \quad (1)$$

Let's assume our streamfunction formulation:

$$u = \frac{\partial \psi}{\partial y} ; \quad v = -\frac{\partial \psi}{\partial x} ; \quad \eta = y \sqrt{\frac{U_\infty}{\nu x}} ; \quad \psi = (U_\infty \nu x)^{1/2} f \quad (2)$$

We will need the following quantities to help us:

$$\frac{\partial \eta}{\partial y} = \frac{\partial}{\partial y} \left(y \sqrt{\frac{U_\infty}{\nu x}} \right) = \sqrt{\frac{U_\infty}{\nu x}} \Rightarrow \boxed{\frac{\partial \eta}{\partial y} = \sqrt{\frac{U_\infty}{\nu x}}} \quad (3)$$

$$\begin{aligned} \frac{\partial \eta}{\partial x} &= \frac{\partial}{\partial x} \left(y \sqrt{\frac{U_\infty}{\nu x}} \right) \Rightarrow U_\infty = x^m \\ &= \frac{\partial}{\partial x} \left(y \sqrt{\frac{x^m}{\nu x}} \right) = \frac{y}{\sqrt{\nu}} \frac{\partial}{\partial x} \sqrt{x^{m-1}} \\ &= \frac{y}{\sqrt{\nu}} \frac{1}{2} (m-1) x^{\frac{1}{2}(m-1)-1} \\ &= \frac{y}{\sqrt{\nu}} \frac{1}{2} \frac{m-1}{x} \sqrt{\frac{x^m}{x}} \\ &= y \frac{(m-1)}{2x} \sqrt{\frac{U_\infty}{\nu x}} \end{aligned}$$

$$\boxed{\frac{\partial \eta}{\partial x} = \frac{(m-1)}{2x} \eta} \quad (4)$$

$$\text{Also remember that } \frac{u}{U_\infty} = f' \Rightarrow \boxed{u = U_\infty f'} \quad (5)$$

And from mass conservation, we can solve for v :

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

We know that:

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial}{\partial x} (U_\infty f') = \frac{\partial U_\infty}{\partial x} f' + \frac{\partial f'}{\partial x} U_\infty \\ &= \frac{\partial}{\partial x} (x^m) f' + \frac{\partial f'}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} \cdot U_\infty \\ \therefore \boxed{\frac{\partial u}{\partial x} &= \underbrace{m x^{m-1} f'}_{m U_\infty x^{-1} \text{ since } U_\infty = x^m} + U_\infty f'' \frac{(m-1)}{2x} \eta} \quad (6)\end{aligned}$$

So now we can solve for v :

$$\frac{\partial v}{\partial y} = - \frac{\partial u}{\partial x} \quad (\text{from continuity})$$

$$\frac{\partial v}{\partial y} = - \frac{U_\infty}{x} (m f' + (m-1) f'' \eta \cdot \frac{1}{2})$$

$$v = \int - \frac{U_\infty}{x} (m f' + \frac{(m-1)}{2} f'' \eta) dy$$

$$\text{We know } \eta = y \sqrt{\frac{U_\infty}{\nu x}} \Rightarrow dy = d\eta \cdot \sqrt{\frac{\nu x}{U_\infty}}$$

$$v = - \frac{U_\infty m}{x} \sqrt{\frac{\nu x}{U_\infty}} \int f' d\eta - \frac{U_\infty (m-1)}{2x} \sqrt{\frac{\nu x}{U_\infty}} \int f'' \eta d\eta$$

We've solved this
before using IBP.
Look on pg.

$$v = - \frac{U_\infty m}{x} \sqrt{\frac{\nu x}{U_\infty}} f - \frac{U_\infty (m-1)}{2x} \sqrt{\frac{\nu x}{U_\infty}} (\eta f' - f')$$

Expanding this:

$$v = - \frac{U_\infty m}{x} \sqrt{\frac{\nu x}{U_\infty}} f - \frac{U_\infty m}{2x} \sqrt{\frac{\nu x}{U_\infty}} (\eta f' - f) + \frac{U_\infty}{2x} \sqrt{\frac{\nu x}{U_\infty}} (\eta f' - f)$$

One more step of expansion:

$$V = -\frac{U_\infty}{x} \sqrt{\frac{Ux'}{U_\infty}} \cdot \left[mf + \frac{m\eta}{2} f' - \frac{mf'}{2} - \frac{\eta f'}{2} + \frac{f}{2} \right]$$

$$V = -\frac{U_\infty}{x} \sqrt{\frac{Ux'}{U_\infty}} \cdot \left[\frac{mf}{2} + \frac{m\eta}{2} f' - \frac{\eta f'}{2} + \frac{f}{2} \right] \quad (7)$$

Now it becomes easy. We just have to plug & chug!
Let's solve the viscous term (Right hand side)

$$\nu \frac{\partial^2 u}{\partial y^2} = \nu \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} (u) \right) \Rightarrow u = U_\infty f'$$

$$= \nu \frac{\partial}{\partial y} \left(\frac{\partial}{\partial \eta} (U_\infty f') \frac{\partial \eta}{\partial y} \right)$$

$$= \nu \frac{\partial^2}{\partial \eta^2} (U_\infty f') \left(\frac{\partial \eta}{\partial y} \right)^2$$

$$= \cancel{\nu} U_\infty f''' \left(\frac{U_\infty}{\cancel{\nu} x} \right)$$

$$\therefore \boxed{\nu \frac{\partial^2 u}{\partial y^2} = \frac{U_\infty^2 f'''}{x}} \quad (8)$$

Let's do the first inertial term:

$$u \frac{\partial u}{\partial x} \Rightarrow u = U_\infty f', \quad \frac{\partial u}{\partial x} \Rightarrow \text{eqn. (6)}$$

$$= U_\infty f' \left(\frac{m U_\infty}{x} f' + U_\infty f'' \frac{(m-1)}{2x} \eta \right)$$

$$u \frac{\partial u}{\partial x} = \frac{U_\infty^2}{x} \left(m(f')^2 + \frac{m}{2} \eta f' f'' - \frac{1}{2} \eta f' f'' \right) \quad (9)$$

Now for the second inertial term:

$$v \frac{\partial u}{\partial y} \Rightarrow v \Rightarrow \text{eqn. (7)}, \quad \frac{\partial u}{\partial y} = \frac{\partial u}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \stackrel{\text{Eqn. (3)}}{=} = \frac{2(U_\infty f')}{2\eta} \cdot \sqrt{\frac{U_\infty}{\nu x}} \\ = U_\infty f'' \sqrt{\frac{U_\infty}{\nu x}}$$

$$v \frac{\partial u}{\partial y} = -\frac{U_\infty}{2x} \sqrt{\frac{\nu x}{U_\infty}} \left[mf + m\eta f' - \eta f' + f \right] \cdot U_\infty f'' \sqrt{\frac{U_\infty}{\nu x}}$$

$$\boxed{v \frac{\partial u}{\partial y} = \frac{U_\infty^2}{2x} (\eta f' f'' - m f f'' - m \eta f' f'' - f f'')} \quad (10)$$

Putting (8), (9), & (10) together : (9) + (10) = (8) + $\frac{U_\infty^2 m}{x}$
 $u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = u \frac{\partial^2 u}{\partial y^2} + \frac{U_\infty^2 m}{x}$

$$\frac{U_\infty^2}{x} \left(m(f')^2 + \frac{m}{2} \eta f' f'' - \frac{1}{2} \eta f' f'' + \frac{1}{2} \eta f' f'' - \frac{m}{2} f f'' - \frac{1}{2} m \eta f' f'' - \frac{1}{2} f f'' \right) = \frac{U_\infty^2}{x} (m + f''')$$

$$m(f')^2 - \frac{1}{2}(m+1) f f'' = m + f'''$$

$$\boxed{f''' + \frac{1}{2}(m+1) f f'' + m(1 - (f')^2) = 0}$$

↳ Falkner - Skan Wedge Flow Momentum Equation.

Now you try the energy eqn!

Note our boundary conditions remain the same:

$$f(0) = 0 ; f'(0) = 0 ; f'(\infty) = 1$$

Note, the solution can be found numerically and is usually tabulated.

Typically for flow problems we need to solve for shear (τ)

$$\begin{aligned} \tau &= \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} \Rightarrow u = f' U_\infty \Rightarrow \sqrt{\frac{U_\infty}{\nu x}} \\ &= \mu U_\infty \left. \frac{\partial f'}{\partial y} \right|_{y=0} = \mu U_\infty \left[\frac{\partial f'}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \right]_{y=0} \end{aligned}$$

Simplifying: (see page 51 for a similar simplification)

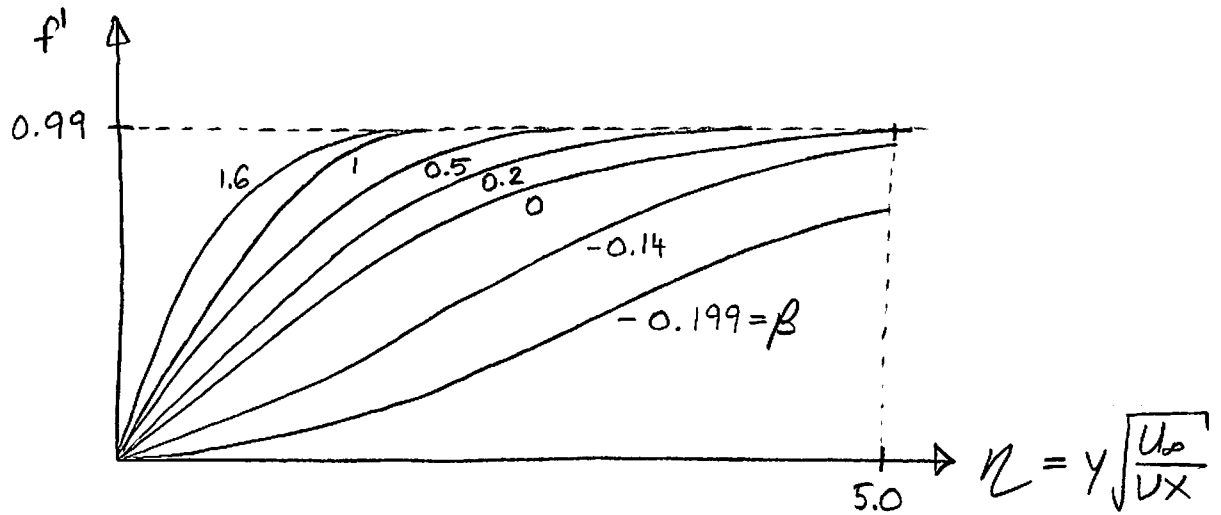
$$C_{f,x} = \frac{\tau}{\frac{1}{2} \rho U_\infty^2} = \frac{2 f''(0)}{Re_x^{1/2}} \Rightarrow \text{Note, } Re_x = \frac{C x^{m+1}}{\nu}$$

↳ Don't forget this!

Our tabulated solutions are:

β	m	$f''(0) = \frac{1}{2} C_{f,x} Re_x^{1/2}$
$2\pi = 6.28$	∞	∞
$\pi = 3.14$	1	1.233
$\pi/2 = 1.57$	$1/3$	0.757
$\pi/5 = 0.627$	$1/9$	0.512
0	0	0.332 $\Rightarrow$ flat plate ($\frac{\partial p_x}{\partial x} = 0$)
-0.14	-0.0654	0.164
-0.199	-0.0904	0 (b.l. separation)

If we plot our velocity profile results



From this, we see that for $\beta > 0$, $m > 0$

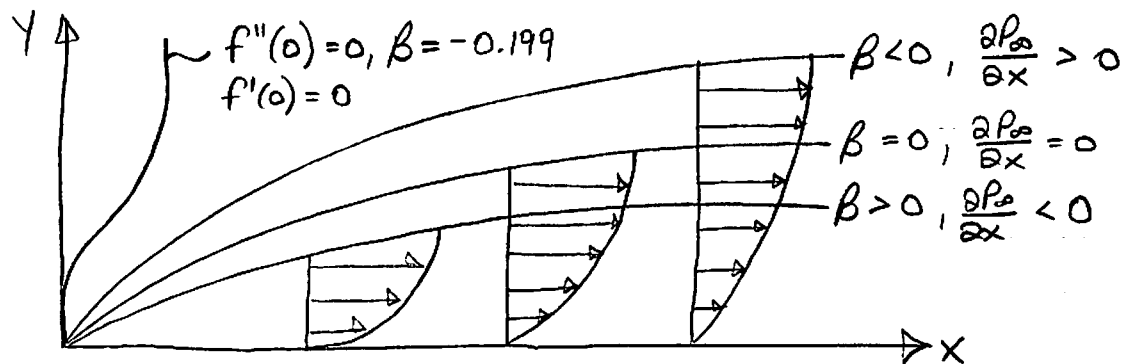
$$\frac{\partial P_\infty}{\partial x} = - \frac{\rho U_\infty^2 m}{x} < 0 \Rightarrow \text{Pressure drops as the flow accelerates}$$

This is a favourable pressure gradient, and our boundary layer gets thinner with larger x -momentum near the wall.

For $\beta < 0$, $m < 0$

$$\frac{\partial P_\infty}{\partial x} = - \frac{\rho U_\infty^2 m}{x} > 0 \Rightarrow \text{Adverse pressure gradient}$$

We see that at $\beta = -0.199$, we have $f''(0) = 0$, so this is called the b.l. separation point



Note, these solutions are called the Fallner-Skan solutions (1931)

Heat Transfer (Fallner - Skan $\Rightarrow$ Wedge Flow)

Our energy equation is exactly the same as before. We can also use the same similarity variables (page 53 of notes)

$$\Theta = \frac{T - T_0}{T_\infty - T_0}$$

$$u \frac{\partial \Theta}{\partial x} + v \frac{\partial \Theta}{\partial y} = \alpha \frac{\partial^2 \Theta}{\partial y^2} \Rightarrow \eta = y \sqrt{\frac{U_\infty}{\nu x}}, \quad f' = \frac{u}{U_\infty}$$

Remember the following:

$$\frac{\partial \Theta}{\partial x} = \frac{\partial \Theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial x} \Rightarrow \frac{\partial \eta}{\partial x} = \frac{\partial}{\partial x} \left(y \sqrt{\frac{C x^m}{\nu x}} \right)$$

$$\frac{\partial \Theta}{\partial y} = \frac{\partial \Theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial y} \Rightarrow \frac{\partial \eta}{\partial y} = \sqrt{\frac{C x^m}{\nu x}}$$

$$\frac{\partial^2 \Theta}{\partial y^2} = \frac{\partial^2 \Theta}{\partial \eta^2} \cdot \left(\frac{\partial \eta}{\partial y} \right)^2 = \frac{\partial^2 \Theta}{\partial \eta^2} \cdot \frac{C x^m}{\nu x}$$

Back substituting & doing simplification, we obtain:

$$\boxed{\Theta'' + \frac{1}{2} Pr(m+1) f \Theta' = 0} \quad (1)$$

This is similar to before except that Pr is replaced with $Pr(m+1)$. Our b.c.'s are identical:

$$\Theta(0) = 0; \quad \Theta(\infty) = 1$$

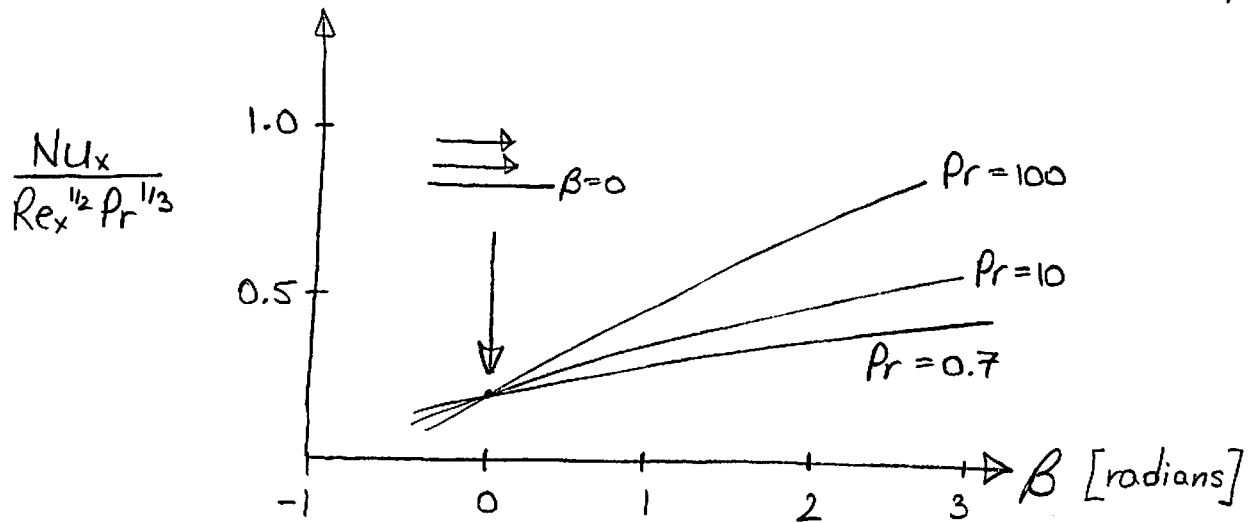
Eckert numerically integrated equation (1) & obtained (for $Pr=1$)

β	m	$Nu_x / Re_x^{1/2}$
-0.512	-0.0753	0.272
0	0	0.332
$\pi/5$	$1/9$	0.378
$\pi/2$	$1/3$	0.440
π	1	0.570

$$Nu = \frac{hx}{k}$$

$\Rightarrow$ Table 2.3 of Bejan [1963]

There is a good way to summarize our results graphically



For any given constant Pr , we can see that:

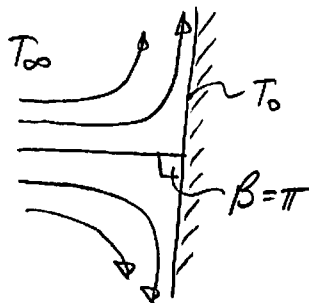
$$Nu_x Re_x^{-1/2} = \text{constant}$$

$$\frac{hx}{k} \left(\frac{x U_\infty}{U} \right)^{-1/2} = \text{constant} \Rightarrow h = \frac{(\text{const.}) k x^{-1/2} U_\infty^{1/2}}{U^{1/2}}$$

But for our problem, $U_\infty = Cx^m$

$$\boxed{h = \frac{(\text{const}) k}{U^{1/2}} x^{(m-1)/2}} \Rightarrow \text{heat transfer coefficient (local)}$$

Note a special case here, for $m=1$, $\beta=\pi$



$$h = \frac{(\text{const}) k}{U^{1/2}} x^{(1-1)/2} = \frac{(\text{const}) k}{U^{1/2}} = \text{CONSTANT}$$

since $h_x = \text{constant}$, this implies that $S_T = \text{const.}$ for $m=1$. Also, $S = \text{constant}$ for this case.

We can say something about the average heat transfer coeff ($\bar{h}$)